

Solving the generalized indirect Fourier transformation (GIFT) by Boltzmann simplex simulated annealing (BSSA)

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The structure of colloidal particles can be studied with small-angle X-ray and neutron scattering (SAXS and SANS). In the case of randomly oriented systems, the indirect Fourier transformation (IFT) is a well established technique for the calculation of model-free real-space information. Interaction leads to an overlap of inter- and intraparticle scattering effects, preventing most detailed interpretations. The recently developed generalized indirect Fourier transformation (GIFT) technique allows these effects to be separated by assuming various models for the interaction, *i.e.* the so-called structure factors. The different analytical behaviour of these structure factors from that of the form factors, describing the intraparticle scattering, allows this separation. The mean-deviation surface is defined by the quality of the fit for different parameter sets of the structure factor. Its global minimum represents the solution. The former non-linear least-squares approach has proved to be inefficient and not very reliable. In this paper, the incorporation of the completely different Boltzmann simplex simulated annealing (BSSA) algorithm for finding the global minimum of the hypersurface is presented. This new method increases not only the calculation speed but also the reliability of the evaluation.

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1. Introduction

Size, shape and internal structure of colloidal particles can be revealed by small-angle X-ray and neutron scattering experiments (Guinier & Fournet, 1955; Glatter & Kratky, 1982; Feigin & Svergun, 1987; Brumberger, 1995). These experiments are performed by measuring the intensity of scattered photons or neutrons for small angles θ , which can be expressed by the modulus of the wavelength-independent scattering vector $\mathbf{q}$, resulting in the so-called scattering curve $I(q)$:

$$|\mathbf{q}| = (4\pi/\lambda) \sin \theta. \quad (1)$$

The scattering curve is related on the one hand to the ensemble-averaged form of the particles, the so-called form factor $P(q)$, and on the other hand to the interparticle ordering, *i.e.* the structure factor $S(q)$. This ordering is a consequence of the interaction potential of the particle, which is also interrelated with its shape.

As long as interaction can be neglected, which is true for diluted and uncharged systems, there is no necessity to take the structure factor into account. In the case of monodisperse homogeneous spherical particles, the total scattering intensity can be expressed as

$$I(q) = N P(q) S(q), \quad (2)$$

where N is the number of particles. Such a separation cannot be performed for polydisperse systems, but similar expressions

can be found for such systems (Kotlarchyk & Chen, 1983; Pedersen, 1994; Weyerich *et al.*, 1999).

One possibility to obtain information about size, shape and internal structure of the particles is the indirect Fourier transformation (IFT) technique (Glatter, 1977), resulting in the pair density distribution function, PDDF. This calculation is performed completely model-free and requires only a rough estimate of the maximum dimension of the particle. The PDDF is approximated by a set of cubic spline functions and the Fourier transformed splines are fitted to the scattering function by a simple linear least-squares routine. The technique works well for diluted and most concentrated systems, but in the case of strong interactions, it is nearly impossible to interpret the resulting PDDF function.

The influence of the interaction effects can be computed by simultaneous calculation of the form factor and the structure factor, leading to a PDDF limited to intraparticle contributions. This can be achieved by the generalized indirect Fourier transformation (GIFT) technique (Brunner-Popela & Glatter, 1997). The main difference from the IFT is that the Fourier transformed splines are multiplied with a parameterized structure factor prior to fitting to the measured scattering curve. This multiplication inhibits the use of a simple linear least-squares fitting algorithm. Brunner-Popela & Glatter (1997) tried to solve this problem by using various non-linear least-squares approaches, such as gradient search, Marquart algorithm and finally linearization of the fit function. These

algorithms worked well with simulated data, but failed to solve most experimental scattering data and gave hardly reproducible results.

In the current paper, we present a completely different and much more efficient approach to solve this optimization problem.

2. Theory

Before going into details, a rough overview of the GIFT method may be helpful. Starting in real space, the PDDF is approximated by a linear combination of a finite number of cubic B-splines in the subspace $0 \leq r \leq D_{\max}$, where $D_{\max}$ is the selectable maximum dimension. The expansion coefficients of the splines are the unknowns.

The form factor $P(q)$ is the Fourier transform of the PDDF. Owing to the linearity of the Fourier transformation, $P(q)$ can be described by a combination of the transformed splines, with the same expansion coefficients as in real space. A fit of the transformed splines to the measured scattering curve including a stabilization criterion gives the best values of the coefficients. Additional linear transformations can be performed to take smearing effects into account (Glatter, 1977).

In the generalized indirect Fourier transformation, there is an expansion by a term for the structure factor according to equation (2), resulting in

$$I(q) = P(q) S(q, \mathbf{d}_N) = \left[\sum_{v=1}^n c_v \psi_v(q) \right] S(q, \mathbf{d}_N), \quad (3)$$

where c_v are the expansion coefficients, ψ are the transformed splines, n is the total number of splines and $\mathbf{d}_N$ is a vector containing the N structure-factor parameters.

The product of the expansion coefficients with the structure factor, which is determined by the unknown parameters $\mathbf{d}_N$, renders the approximation non-linear and excludes the use of any simple linear least-squares algorithm. In order to solve this highly non-linear problem, a separation into a linear calculation of the expansion coefficients with a fixed parameter set of the structure factor and a non-linear optimization of the structure factor can be performed. This is performed in iterative steps (Fig. 1) until the best mean deviation MD is found, according to

$$MD = \sum_{i=1}^m \{ [I_{\exp}(q) - \tilde{I}(q_i, \mathbf{c}, \mathbf{d}_N)] / \sigma(q_i) \}^2, \quad (4)$$

with the experimental scattering curve $I_{\exp}(q)$ and the fit $\tilde{I}(q)$.

The iteration step to calculate the expansion coefficients with a structure factor with a fixed set of parameters is analogous to the IFT technique. Detailed explanations can be found in an article by Glatter (1977).

A very efficient way to solve optimization problems of continuous multidimensional hyperspaces has been published by Press *et al.* (1992). This Boltzmann simplex simulated annealing (BSSA) algorithm has proved to be very well suited for solving the other iteration step, namely the search for the best $S(q)$ parameters.

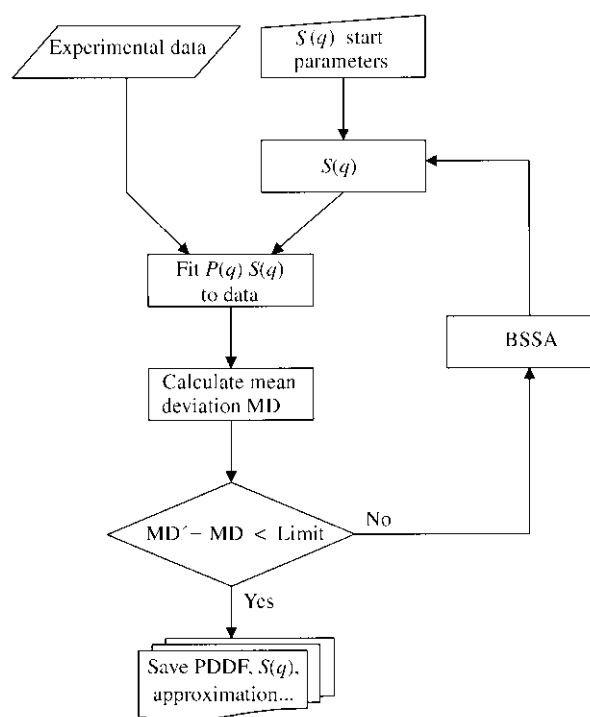


Figure 1

Programming flowchart of the generalized indirect Fourier transformation technique using the Boltzmann simplex simulated-annealing algorithm. MD' is the mean deviation of the previous iteration step; MD is the mean deviation of the current iteration step.

The BSSA algorithm is a modification of the downhill simplex (DS) algorithm (Nelder & Mead, 1965), which can be used to optimize a local minimum of an arbitrary function. In order to reach this goal, we consider the optimization of the minimum of the MD hypersurface with the vector $\mathbf{d}_N$ specifying the structure factor. Firstly, a set of $N + 1$ initial parameter-set vectors $\mathbf{d}_N$ in the vicinity of the local minimum has to be specified. Each vector is referred to as a vertex; the set of $N + 1$ vertices is called the simplex, which defines an N -dimensional polyhedron enclosing an N -dimensional hypervolume. At each vertex, the MD is calculated. In the evolution of the simplex, the first step that the algorithm tries is a reflection: the highest MD vertex is reflected through the hyperplane formed by the other N vertices conserving the hypervolume of the simplex. The following step can either be a reflection expansion or contraction depending on the MD found after the reflection. If the reflection move gave a lower resulting MD than the lowest MD of all of the previous vertices, the reflection expansion is performed to move rapidly downhill where appropriate. If the result of the reflection gave an MD that was worse than the second highest MD of the previous vertices, a contraction is tried: the highest vertex is moved along one dimension towards the lowest MD vertex. If both possibilities (reflection expansion and contraction) failed, a multiple contraction is carried out. This move shifts the highest MD vertex along all the coordinates towards the lowest MD vertex. After the preceding series of move attempts has been made, the process is repeated until the

convergence criterion is met or the maximum number of iterations is reached.

The pure DS algorithm will converge to an accessible local MD minimum, but will not be a reliable tool to find the global MD minimum in the presence of many local minima.

We have implemented the BSSA algorithm as presented by Press *et al.* in their *amebsa* routine, with some modifications required by the GIFT technique. The main difference between the DS and the BSSA is the introduction of 'temperature'-dependent random fluctuations into the MD values at each step (reflection, reflection expansion, contraction): a positive logarithmically distributed random variable z , proportional to a so-called 'temperature' T corresponding to a Boltzmann probability distribution, is added to the MD of every vertex of the simplex, according to

$$MD' = MD(\mathbf{d}_N) + kT \log(z), \quad (5)$$

where k is the Boltzmann constant. A similar random number is subtracted from the MD of every new vertex that is tried as a replacement vertex. Like the Metropolis algorithm (Metropolis *et al.*, 1953), this method always accepts a true downhill step, but sometimes accepts an uphill step as well. As soon as $T = 0$, the BSSA algorithm degenerates to the DS procedure and converges to a local minimum.

The so-called 'temperature' is reduced according to an annealing schedule in an iterative procedure. At a finite value of T , the simplex expands to a scale that is approximately the size of the region accessible at this 'temperature', and then carries out a Brownian motion within this region, sampling new approximately random vertices. If the temperature is reduced sufficiently slowly (annealing), there is a high probability towards the lowest MD minimum encountered.

This annealing schedule is a crucial part of the application of this BSSA algorithm. In our case, after intensive testing of different cooling schemes, we reduce T to $(1 - \varepsilon)T$ after j moves, where $\varepsilon = 0.2$ gave very reproducible results in almost any case and $20 \leq j \leq 40$, depending on N (Fig. 2). The starting value of T is determined by the MD of the initial vertices, which gives a good scaling for different hypersurfaces.

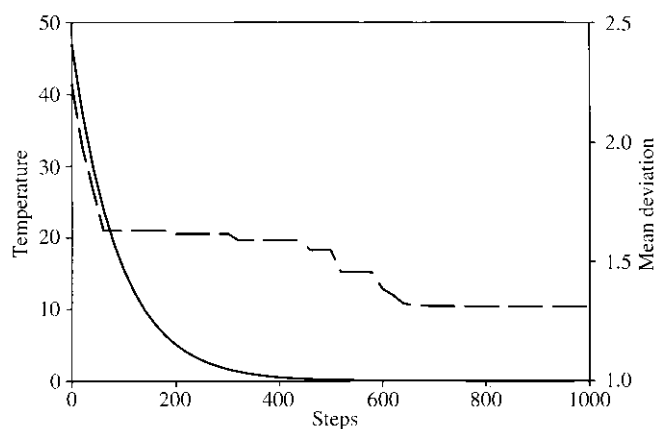


Figure 2
Development of the annealing temperature (solid) and corresponding best mean deviation (dashed) during the calculation.

Because the new vertices are chosen randomly, the minimum search has to be limited to physically possible parameter sets. Therefore, any step into the region of unphysical parameters is charged by a mean deviation of infinity (e.g. negative interaction radii, negative volume fractions). The optimization procedure between the limits is not changed.

3. Results and discussion

In order to demonstrate the improvements and advantages of the GIFT technique using the BSSA algorithm, we shall present some examples of experimental results. The sample used for all evaluations was 6.77% (w/w) $C_{12}E_6$ (hexaoxyethylene glycol mono-*n*-dodecyl ether) in D_2O at 276 K. The measurements have been performed at the SANS instrument D22 at ILL in Grenoble. A detailed description and discussion of the samples will be given by Glatter *et al.* (2000).

The annealing schedule leads to typical development of the structure-factor parameters during evaluation (Fig. 3). In the high-temperature regime, moves to even high mean deviations are possible, leading to strongly changing parameters for the structure-factor with access to the whole parameter hypersurface. With decreasing temperature (corresponding to the annealing schedule), the oscillations become weaker and remain stable in the vicinity of the global minimum. This behaviour makes the use of a second regularization for the effective structure factor as introduced by Weyerich *et al.* (1999) obsolete.

As shown in Fig. 4, the GIFT technique with the BSSA algorithm converges to the global minimum of the parameter hypersurface. The formerly used linearization of the fit function does not give results of such high reliability. It is obvious that Fig. 4 represents quite a small part of the hyperspace. Nevertheless, it has to be explained why the linearization of the fit function is not found to be converging into the global minimum. On the one hand, it was necessary to damp the parameter changes during iteration in order to prevent uncontrolled jumps in the hypersurface. On the other hand, as

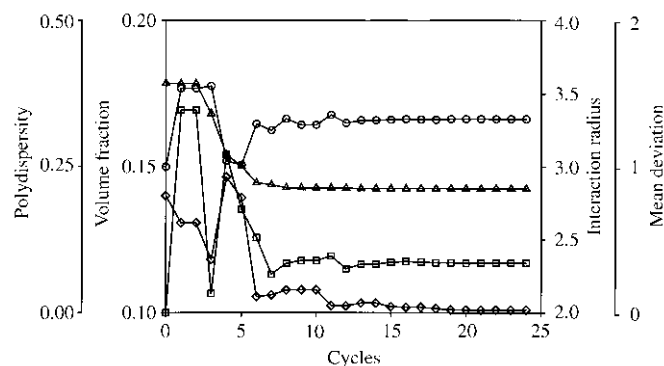


Figure 3
Evolution of the structure-factor parameters and the corresponding mean deviation during the calculation. Squares: volume fraction. Diamonds: polydispersity. Circles: interaction radius. Triangles: mean deviation.

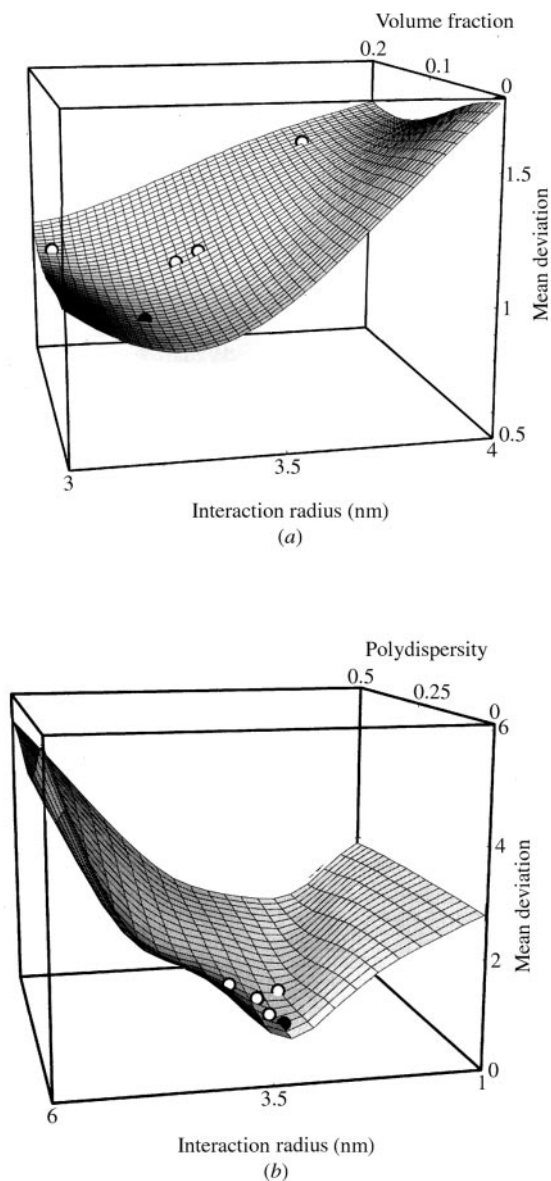


Figure 4

Mean-deviation hypersurfaces of structure-factor parameters. (a) Surface for a polydispersity of 0.5%; (b) surface for a volume fraction of 13.3%. The filled circles represent the solution found by the BSSA algorithm, while the open circles mark solutions of the linearization of the fit function.

soon as the slope of the surface becomes small, the damping makes the convergence inefficient; the iteration was usually stopped after a fixed number of steps to limit the necessary computing time [a detailed description has been presented by Brunner-Popela & Glatter (1997)]. As a model for the structure factor, the averaged hard-sphere structure factor introduced by Brunner-Popela & Glatter (1997) was used. In this case, the number of parameters (N) for the structure factor is 3 (volume fraction ϕ , hard-sphere radius R_{HS} and polydispersity parameter μ). A deeper insight into the physics of the different hard-sphere structure factors can be gained by consulting the paper by Weyerich *et al.* (1999). The parameters corresponding to the global minimum of the hypersurface

result in the structure factor shown in Fig. 5(a). The fact that the former algorithm (linearization of the fit function) does not reach the hypersurface minimum leads to a different shape of the structure factor. Hence, the resulting PDDFs (Fig. 5b) show different structures.

In the case of using the BSSA algorithm, the shape of the PDDF indicates homogeneous spheres with a maximum dimension of 6.5 nm, whereas the linearization of the fit function leads to a PDDF which can be interpreted as a slightly elongated structure with a maximum dimension of 9.5 nm. The D_{max} value chosen for the evaluation of the data was in both cases 10 nm. The fact that in the case of using the BSSA algorithm the maximum dimension of the particle (as shown in Fig. 5b) is found to be smaller is a proof for the stability of the model-free evaluation method and shows that the information about the actual maximum dimension is not required to restore, uniquely, the PDDF and $S(q)$. Other evidence for the correctness of the result obtained is the fact that at low concentrations (0.25% w/w), C_{12}E_6 in D_2O at 276 K forms micelles of similar size and shape (Glatter *et al.*, 2000).

It is apparent from these results that it is necessary to find the global minimum of the mean-deviation hypersurface in order to find correct solutions. As long as we deal with

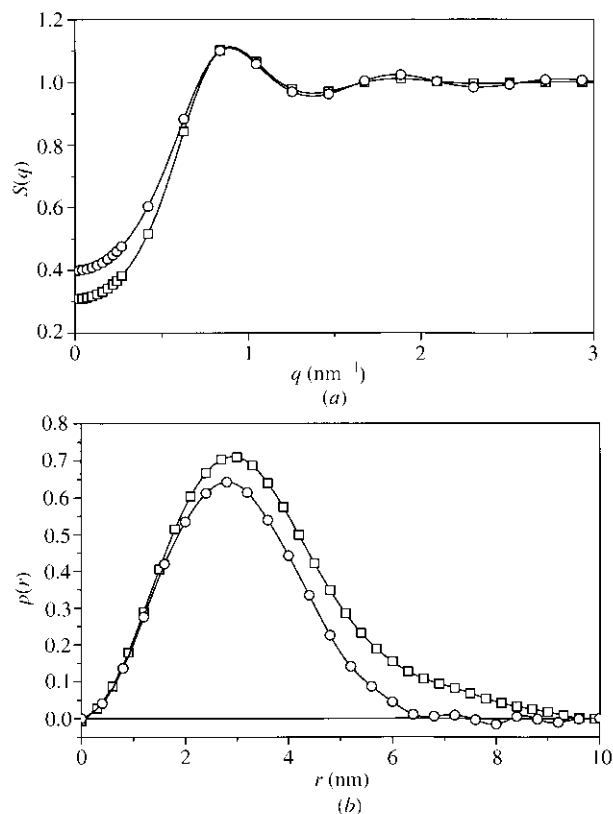


Figure 5

(a) Averaged structure factor of a 6.77% C_{12}E_6 solution at 276 K in D_2O . Solutions were obtained by the linearization of the fit function (squares) and by the BSSA algorithm (circles). (b) PDDFs of a 6.77% C_{12}E_6 solution at 276 K in D_2O . Solutions were obtained by the linearization of the fit function (squares) and by the BSSA algorithm (circles).

simulated data, the reliability of the algorithm with the linearization of the fit function is sufficient (Brunner-Popela & Glatter, 1997). With experimental data the situation is different; after thorough testing, the BSSA algorithm has proven itself able to find the global minimum in almost all cases in even shorter calculation times (typically in the range of 10–30 minutes on a Pentium 200 PC).

On applying the GIFT method, one still has to consider that the calculation of the structure factor is performed by using a model. As soon as the resulting parameters for the structure factor become physically incorrect, the limits of the method are exceeded and the calculated PDDF function is no longer correct. Therefore, the possibility remains to use a more accurate structure-factor model for the description of the scattering properties of the sample.

Within the range of physically possible values, the method has been proven to be stable and can be used for a large variety of different structure factors, such as hard-sphere structure factors [averaged, effective structure factor (Weyerich *et al.*, 1999)] and charged structure factors [rescaled mean spherical approximation, hyper-netted chain, Rogers Young (Fritz *et al.*, 2000)].

Information about the availability of the software can be found at <http://physchem.kfunigraz.ac.at/sm>.

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